

Pricing invoices against a number nobody can see

How we turned a 60-second pricing game into a measured payoff function, a market memory, and a learning loop — and what the data taught us along the way.

1st

×3 in v2's first six rounds

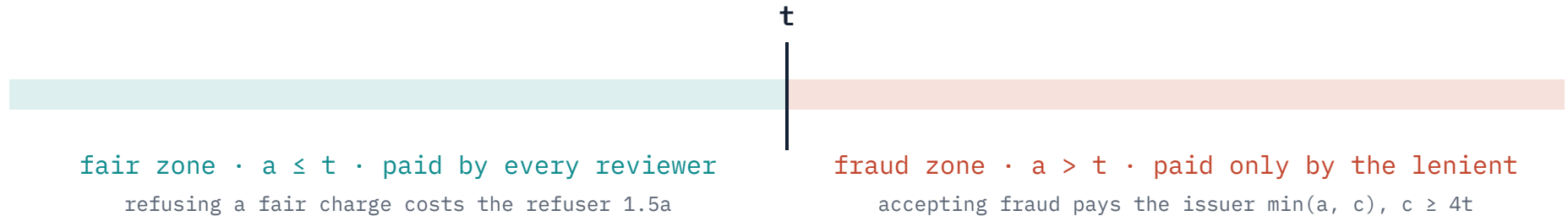
0.08

vs 0.41 · memory vs LLM log-error

5–16 s

per round, one call

Every line item has a secret fair value t . You set a charge a and a limit b .



Issuer: the payoff is a step

$$R(a) = a \cdot P(t \geq a) + 0.5 \cdot t \cdot P(t < a)$$

Measured on the field: revenue per opponent above t is flat at $0.48 - 0.53 \cdot t$ for every overcharge ratio from 1.0 to 5. Crossing t forfeits half immediately and nothing more is gained beyond it.

Reviewer: accept iff $P(\text{fair}) > \frac{2}{3}$

accept: cost a · refuse: cost $1.5a \cdot P(\text{fair})$
 $\Rightarrow b = Q(\frac{2}{3})$ of an honest belief

With coverage inside it: $b = \sup\{a : P(\text{cov}) \cdot S(a) \geq \frac{2}{3}\}$. Everything hinges on the belief being honest — not on the quantile being tuned.

Six things the data said that we did not expect

63

rd pct

t is exogenous. It sits at the 63rd percentile of the field's charges (IQR 0.33–0.86). There is a real number to estimate; the market is not the answer key.

0.08

vs 0.41

Memory beats the model. For recurring line items, past proven brackets predict t with MAD 0.08–0.10 in log space; the LLM 0.36–0.41 on the same items. Averaging the two is *worse* than memory alone.

–0.03

The LLM's interval is uninformative. Its stated q10–q90 has correlation –0.03 with its realised error. σ must come from measured residuals.

0.5 t

Overcharging is flat. The fitted $p_0 \cdot r^{-k}$ curve ($k < 1$) claimed revenue rises forever — a censoring artifact. Refusals reconstructed, it is a flat plateau.

2

items

Whales decide rounds. The best team's entire lead was two items charged at $\sim 2\times$ the field median and still fair (9,585 vs our 4,771; 8,942 vs 5,285).

–116k

Presence beats pricing. Two rounds with no board cost more than all pricing errors of the same 15-round window combined.

One pipeline, two engines, a loop that learns after every round

Extract

decrypt, pypdf, POS + label parser (layout-only, no unit vocabulary)

~0.3 s

Estimate

one OpenAI call; v1 chunks ≥ 10 items in parallel; v2 three samples, per-item median

v1 20–40 s · v2 5–16 s

Price

lognormal belief $\rightarrow$ step-EV charge, $\frac{2}{3}$ -rule limit, whale floors, coverage inside the rules

ms

Submit

first board the moment it exists; later passes overwrite (last write wins)

first board ~15 s

Learn

truth brackets from public payouts; calibration refit; reliability alarm; market memory

after close

Engine v1 — the model prices everything

$t_{\text{low}}/t_{\text{mid}}/t_{\text{high}}$ from the prompt; a 3-segment log-log spline corrects size-dependent bias (β below a pivot, above it, and a whale segment); σ by interval-censored MLE; per-category bias buckets with shrinkage.

Engine v2 — memory first, model second

Per-line memory prior from every past proven bracket, precision-weighted against the LLM; a 1,400-character prompt with anchors inline and `p_covered` instead of a boolean; autoplay launches every round 90 s early.

Autoplay: reads the schedule, never re-plays a logged round, never runs two rounds at once. Presence is the whole job.

The belief is the whole decision — a and b are derived, not tuned

```
ln t | covered ~ N( $\mu$ ,  $\sigma^2$ )
 $\mu$  = precision-weighted( memory  $\mu_M/\sigma_M^2$  , LLM ln q50/ $\sigma_L^2$  )
a = argmax_a P(cov) · [ a · S(a) + 0.5 · E[t ; t < a] ] (capped at Q(0.55))
b = exp(  $\mu + \sigma \cdot \Phi^{-1}(1 - \frac{1}{3} / P(\text{cov}))$  )
whale (q50 ≥ 2k): b ≥ posterior median
```

- **Why pure EV, not mean – $\lambda \cdot \text{sd}$:** a round is a portfolio of 10–20 items × 16 opponents; per-item shading forfeited 21–40 % of provably collectable income. Re-priced regret fell 748,539 → 516,256 (–31 %).
- **Why B_QUANTILE = 1/3 is derived:** the tuned values (0.33 → 0.40) were patches for a mis-centred belief. Centre μ instead (live_shift, reliability-monitored) and the quantile is the number the math gives.
- **Why memory is price, not coverage:** a past refusal predicts the next case's coverage only 40 % of the time. Policies vary per case; market prices do not.

Two-sided counterfactual (the honest test)

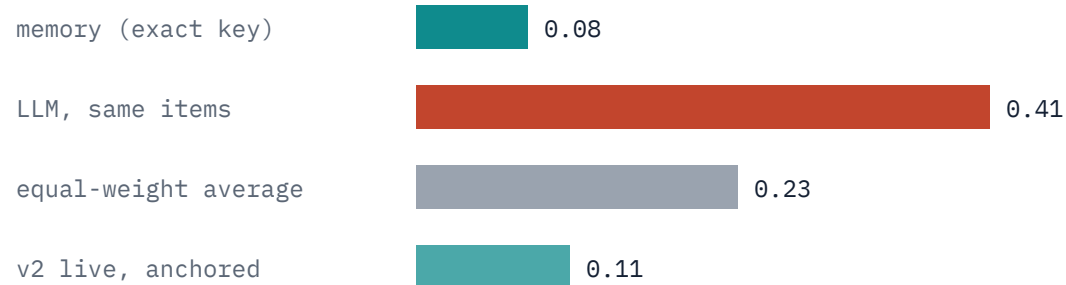
Income from truth brackets, expense against the charges actually billed to us. Shipped board → new pricing, our proven-fraud count unchanged.

ROUND	SHIPPED	NEW
41	–1,405	+49,042
42	–17,816	–17,400
45	+4,120	+18,239

The deviation metric alone would have picked a higher cap — it costs under-charging only. Both sides, or nothing.

The market remembers what it paid. We made that the prior.

Median $|\log \text{error}|$ vs proven t^* , same 26 items (rounds 38–45)



3-word keys: 68 % of items · 60 % of proven value carry a memory anchor

- Precision-weighted, so memory dominates where it exists (σ 0.15 vs 0.6); the LLM carries novel items and whales.
- Anchors go *inline per item* in the prompt — "paid ~€58–86 per line when covered (n=11)" — not one generic history block.
- Indemnity lines (stolen goods, declared values) take memory only as a floor: another case's watch says nothing about this one.

The v2 prompt, in full spirit

For every line: p_{covered} (0–1, calibrated: $p \geq 0.9 \rightarrow 42$ covered / 3 not; $p < 0.1 \rightarrow 0$ / 6), honest q_{10} / q_{50} / q_{90} , a 12-word reason. Policy limits bind. Whales first. 1,400 characters; three parallel samples; first to land is submitted as insurance, the per-item median overwrites it.

207

memory keys after
game 96

15 s

typical first board on
the server

v2 live: three firsts in six rounds – and one round that taught us the most

ROUND	INCOME	EXPENSE	NET	RANK	NOTE
48	52,331	35,884	+16,447	1/17	27 items, first board 22.6 s
49	23,204	14,410	+8,794	1/17	
50	23,892	17,047	+6,845	4/17	
51	45,211	58,084	-12,873	10/17	two novel items, no memory
52	971	4,467	-3,496	15/17	2-item round
53	138,024	55,985	+82,039	1/17	whale charged at the proven max
54	–	–	-48,994		memory poisoning (slide 9)

Backtest, rounds 38–45 (same simulator, real opponents)

	EXPECTED	PESSIMISTIC
v1	162,489	-18,872
v2	170,576	-15,124
eyay (best team, actual)	175,917	–

Where memory covers most items (40, 42, 45) v2 is +24k / +20k / +6k. The two whale rounds carry 80 % of either total and hinge on the unproven upper side of t.

Every bad round produced a number, not an opinion

Truth from public payouts

Each round's feed proves brackets: a charge paid by all is fair ($t \geq a$); a charge refused by all and paid 0 is fraud ($t < a$). Matchup cells tighten them. 47 → 96 truth files.

Calibration with a drift alarm

Interval-censored MLE of bias and σ on current-prompt estimates only; a `live_shift` fitted on live rounds; make `learn` prints the measured $P(t \geq Q(1/3))$ against the $2/3$ the rule needs.

Replay, score, decide

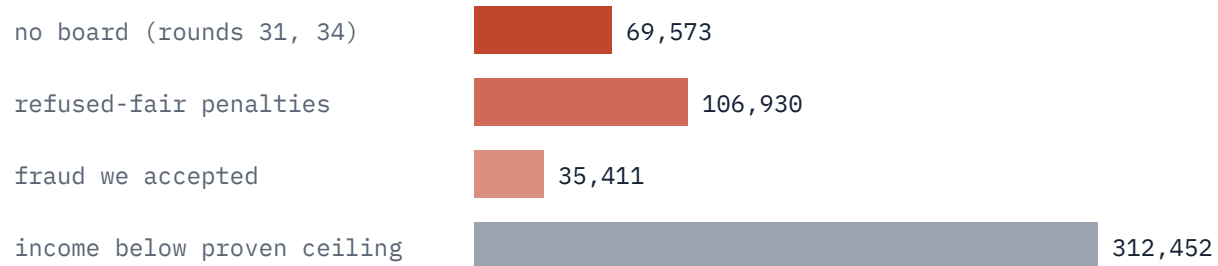
Stored estimates are re-priced in seconds; a change ships when it wins on money in a majority of rounds and on the two-sided counterfactual — never on a one-sided metric.

What the loop caught

- The belief's reliability *flipped sign* between prompt eras — claimed $P(t \geq b)$ 0.60, measured [0.67, 0.76] on the old prompt and [0.43, 0.43] live on the new one. Seven knobs were patching the same residual on different windows.
- Widening σ to the measured residual (≈ 1.15) *loses* money (−80k leave-one-out): the decision-relevant σ is not the residual σ , because the payoff is asymmetric and bounded.
- The "accept everything" idea: a team with a 93 % accept rate broke even and paid 25k in fraud in one round. The winners accept the *right* charges.

Where the money actually went (rounds 30–44: -165k)

EUR lost, by cause



Operational losses (two blank rounds, a Windows path / 7z chain, a 33 s call against a 60 s window) exceeded every pricing error of the same window.

The one that taught us the most: round 54, -48.6k

Game 7's air-con unit was *refused as not covered* ($t < 80.83$). v2's memory read that ceiling as a **price band of €47–77**; the prompt made memory binding; the model obeyed — $q_{50} = 62$ on a €2,000 unit, copied onto its sibling — and $b = 54$ refused every fair charge on the two biggest items.

Fix (v4): memory carries price from *paid* observations only; refusals never lower coverage; anchors advisory and quantity-aware. Same estimates, math only: **+17 %**; fresh replay turned -48.6k into -10.7k on that round.

Why we succeeded – and why we did not

Succeeded

- The measured-payoff maths paid immediately and measurably: step objective, derived b, coverage inside the rule — -31 % regret re-priced, +49k / +18k on the counterfactual rounds.
- Market memory was the single best idea: 4× the model's precision on recurring lines, three firsts in v2's first six rounds, at a fifth of the latency.
- The learning loop was real — truth, calibration, reliability alarm, replay — so fixes were validated before they went live, and mistakes produced numbers.

Did not

- Operations cost more than modelling: ~120k to missed rounds and toolchain breakage; autoplay came late.
- The LLM's level error on first-sight items (+0.36, MAD 0.8 live) is the residual we never fixed — no constant can, only more memory coverage or a prior for novel items.
- We tuned on windows that moved under us (replay vs live, old vs new prompt) several times before building the live-only monitor.
- Several engines in one working tree across machines: artefact collisions on every pull, and two machines once played the same round.